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4. (a) The table shows the results of an experiment in which zinc and lead powders were added separately to solutions of sodium chloride and iron(II) chloride.

	Sodium chloride solution	Iron(II) chloride solution
zinc	no change	colour change
lead	no change	no change

Use the results to place the four metals in order of reactivity. [1]

Most reactive ....

....

....

Least reactive ....

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(b) A student investigated metal reactivity using a different method.

He added 10.0 g of magnesium powder in 2.0 g portions to 50 cm^3 of zinc chloride solution and recorded the temperature of the mixture after each addition. He repeated the experiment with aluminium powder and again with copper powder.

The results for magnesium powder are shown below.

Mass of magnesium powder added (g)	Temperature (°C)
0	21.0
2.0	23.2
4.0	25.1
6.0	27.5
8.0	29.7
10.0	31.6

The results for aluminium and copper have been plotted on the grid opposite.

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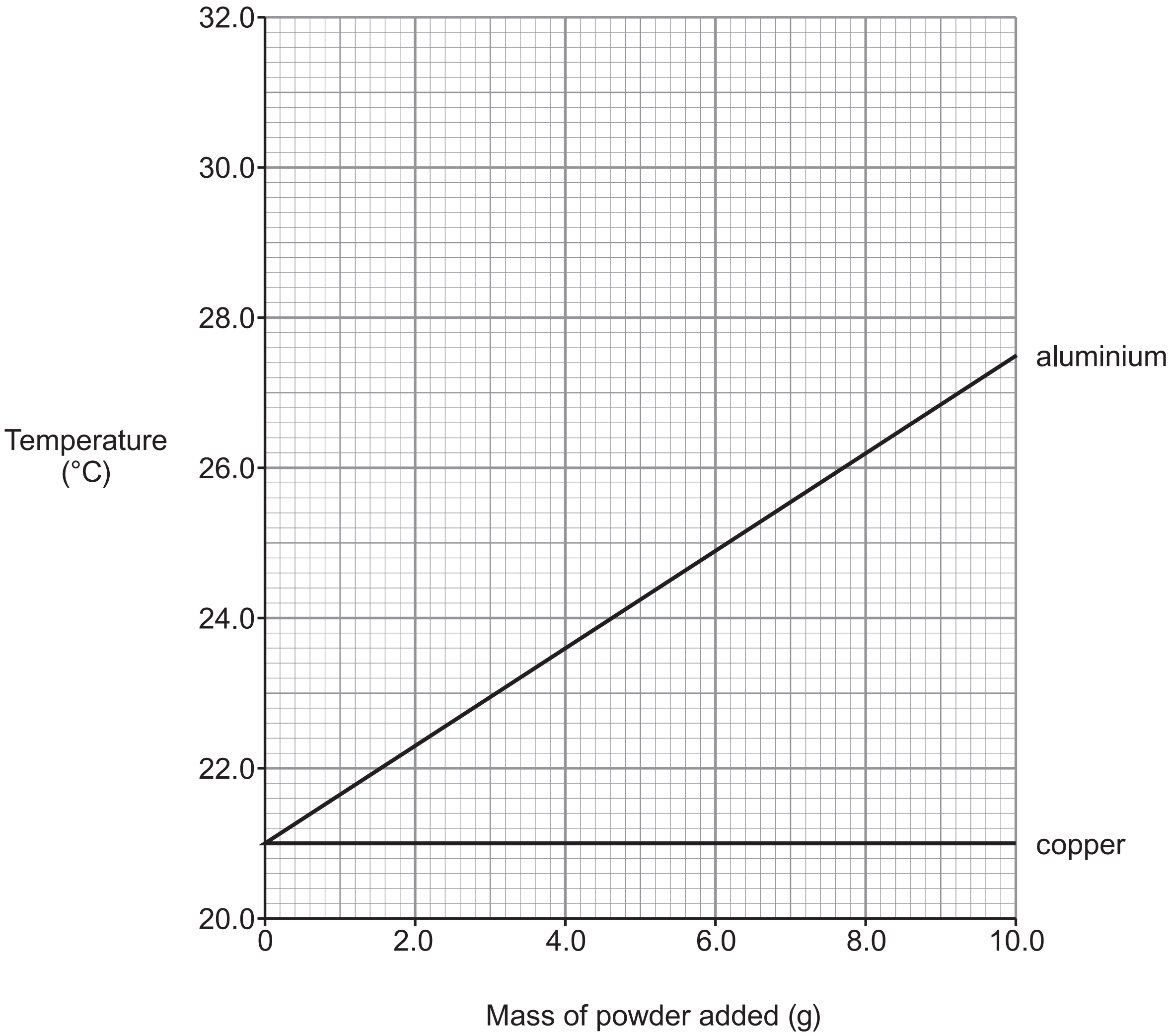
(i) Plot the results for magnesium on the grid. Draw a suitable line. [3]



(ii) State the conclusions that can be drawn about the reactivities of magnesium, aluminium, copper and zinc. Give your reasoning. [3]

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